CHAPTER 1

Tracker

1.1 Pixel detector

The pixel detector comprises four barrel layers and nine discs at each end, with 5 362 modules and 11 806 active chip patches. The inner two barrels use single-chip modules and the outer two use quads. The endcaps combine four inner single-module rings and two outer quad rings, with a larger active aperture at the forward stations. Carbon sandwich staves and discs carry the modules, cooling and local services. This is a preliminary simulation baseline; the mounting joints, common support stiffness and cooling margins require engineering qualification. Figure 1.1 shows the selected layout and its connection to the outer service route.

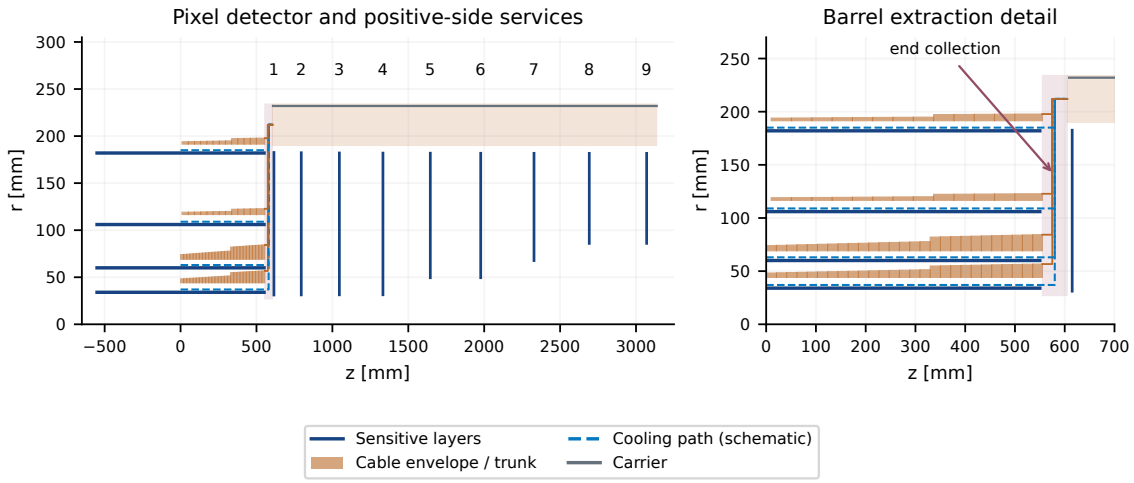


Figure 1.1. Pixel barrel and positive endcap, including local cooling, cable routes and the barrel collection bay. The opposite end is reflected in z.

1.1.1 Modules and readout

Each hybrid module uses a planar n-in-p sensor, ITkPix v2 ASICs and a backside flex. The 400 by 384 matrix has square 50 µm pixels, giving an active area of 20 mm by 19.2 mm and 153 600 channels per chip. The approximate 20 mm by 21 mm die reserves the remaining space for chip periphery.[1] A quad shares one sensor and flex between four active patches, with 614 400 channels. Here a guard is the inactive silicon border excluded from the geometric active map, rather than a specified fabricated guard-ring implant. These borders

and the internal chip seams remain passive silicon. The substrate, active patches and occupied module body are represented separately; their dimensions are given in Table 1.1.

In n-in-p sensors the implants and guard structures face the readout chips; the backside flex carries bias, power and signals. Edge insulation separates the biased sensor from nearby chip surfaces.[2] The flex uses a 25 µm polyimide core and two 12.5 µm coverlays, represented together as 50 µm of polyimide. Two 18 µm copper foils at 50 % coverage and 25 µm epoxy complete its effective material inventory. Sensor and ASIC silicon are each 150 µm thick, separated by a 20 µm bump standoff. Chip peripheries connect to exposed flex pads through the selected 25 µm aluminium bond-wire concept and a module-side termination.

The stack runs from flex copper, polyimide, inner copper and epoxy through sensor, bump standoff, ASIC and graphite contact toward the support. The graphite contact shims are thermally conducting spacers behind each ASIC. Their 0.255 mm thickness fills the nominal distance from the ASIC backs at $w = 0.245$ mm to the body-back plane at $w = 0.5$ mm, where the stave or disc pickup begins. This is a material allocation; the graphite orientation, bond and contact resistance still require specification. Insulation separates the module's serial-power potential from conductive supports. Figure 1.2 combines the module footprints and stack; the occupied body is 1 mm deep. Endcap single bodies are 20.4 mm tangentially by 21.4 mm radially; quad bodies are 43.2 mm by 44.2 mm, with ASIC peripheries at the outer radial edges.

Table 1.1. Physical sensor and active-map dimensions in local (u, v), in mm. Active bounding spans include the quad's internal seams; active area is the sum of the separate patches.

	Single barrel	Quad barrel	Single endcap	Quad endcap
Active bounding span	19.2×20.0	38.6×40.2	20.0×19.2	40.2×38.6
Sensor substrate	20.2×20.4	39.6×40.6	20.2×19.4	40.4×38.8
Guard per outer edge	0.5, 0.2	0.5, 0.2	0.1, 0.1	0.1, 0.1
Active area [mm^2]	384	1536	384	1536

1.1.2 Barrel layout

The four barrel reference radii are 34 mm, 60 mm, 106 mm and 182 mm. Single-chip staves tile the smaller circumferences; quads reduce repeated module edges at larger radii. Adjacent staves alternate radial levels, with the offsets and populations in Table 1.2. Their modules have no azimuthal tilt. The 82 staves carry 3 050 modules and 6 794 chip patches.

Local u is tangential, v follows z, and w points radially outward. The 20 mm active dimension follows z, placing the approximately 1.8 mm chip periphery at the transverse edge. The longitudinal guard and body gap are each 0.20 mm. Single/quad body lengths are 20.4 mm and 40.6 mm, with pitches 20.6 mm and 40.8 mm; neighbouring active edges are separated by 0.60 mm, and quad chip seams are 0.20 mm. The selected *slim longitudinal edge* is the 0.20 mm inactive sensor border along the barrel's z direction. Public slim-edge sensor development motivates this choice; the selected dimensions and operating bias require qualification.[2]

Rows have constant pitch within the 550 mm half-length allocation, with one chip centre at $z = 0$ and a common longitudinal phase between staves. The single-module

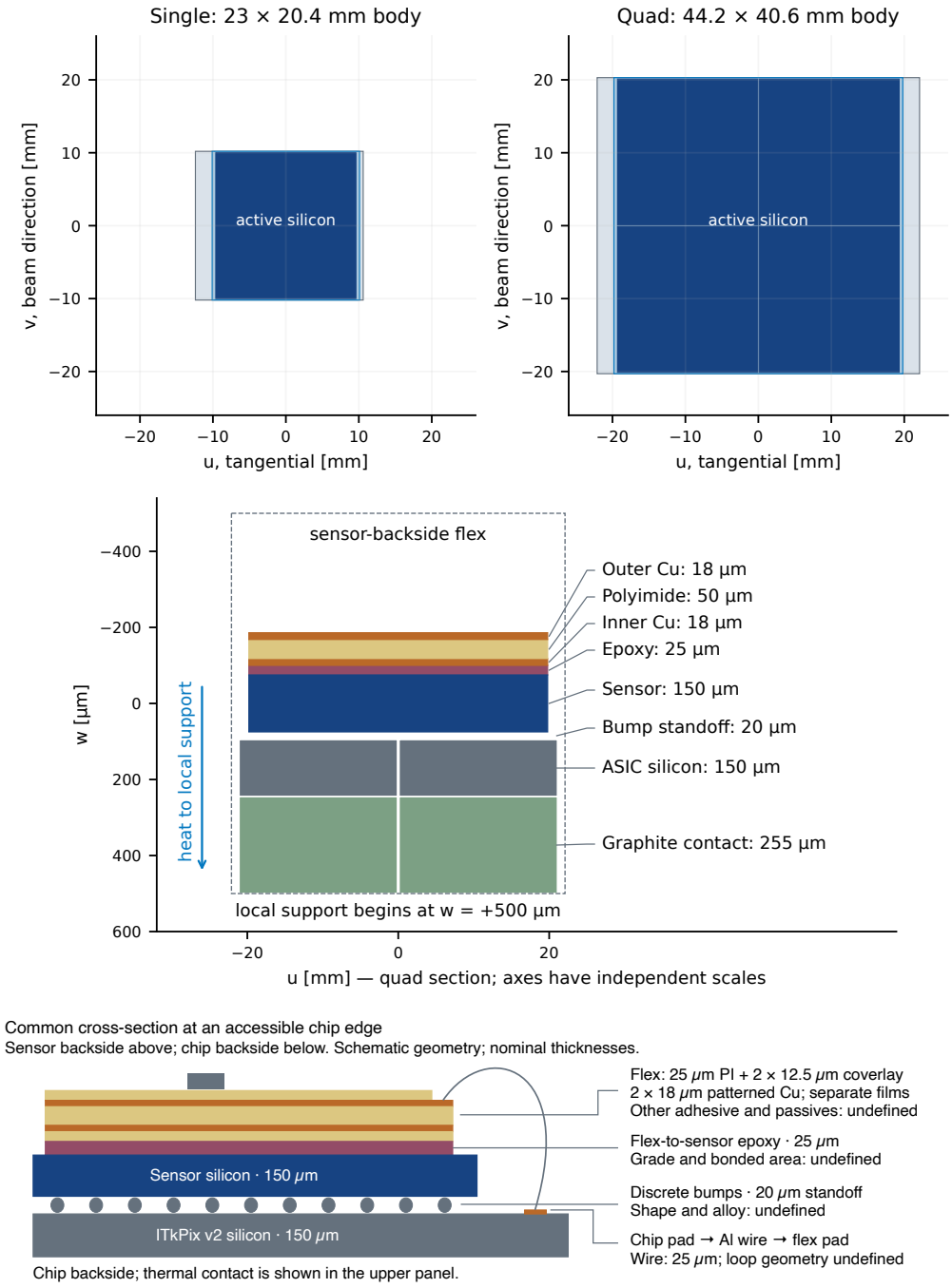


Figure 1.2. Module footprints (top), the quad material stack through one chip row (middle), and schematic bond access (bottom). Active patches, passive silicon and occupied envelopes are distinct. Stack thicknesses use an independent scale; the electrical detail illustrates discrete connections.

body is displaced 0.9 mm tangentially from its active centre. Figure 1.3 shows the layer arrangement and packing.

Table 1.2. Selected barrel population. Radial offsets are rounded display values; the printed precision does not specify manufacturing tolerances. Rows means modules per stave. All dimensions are in mm.

Layer	Radius r	Offset ±	Staves	Rows	Modules
B1	34	2.492	12	53	636
B2	60	1.692	22	53	1166
B3	106	2.778	18	26	468
B4	182	1.877	30	26	780

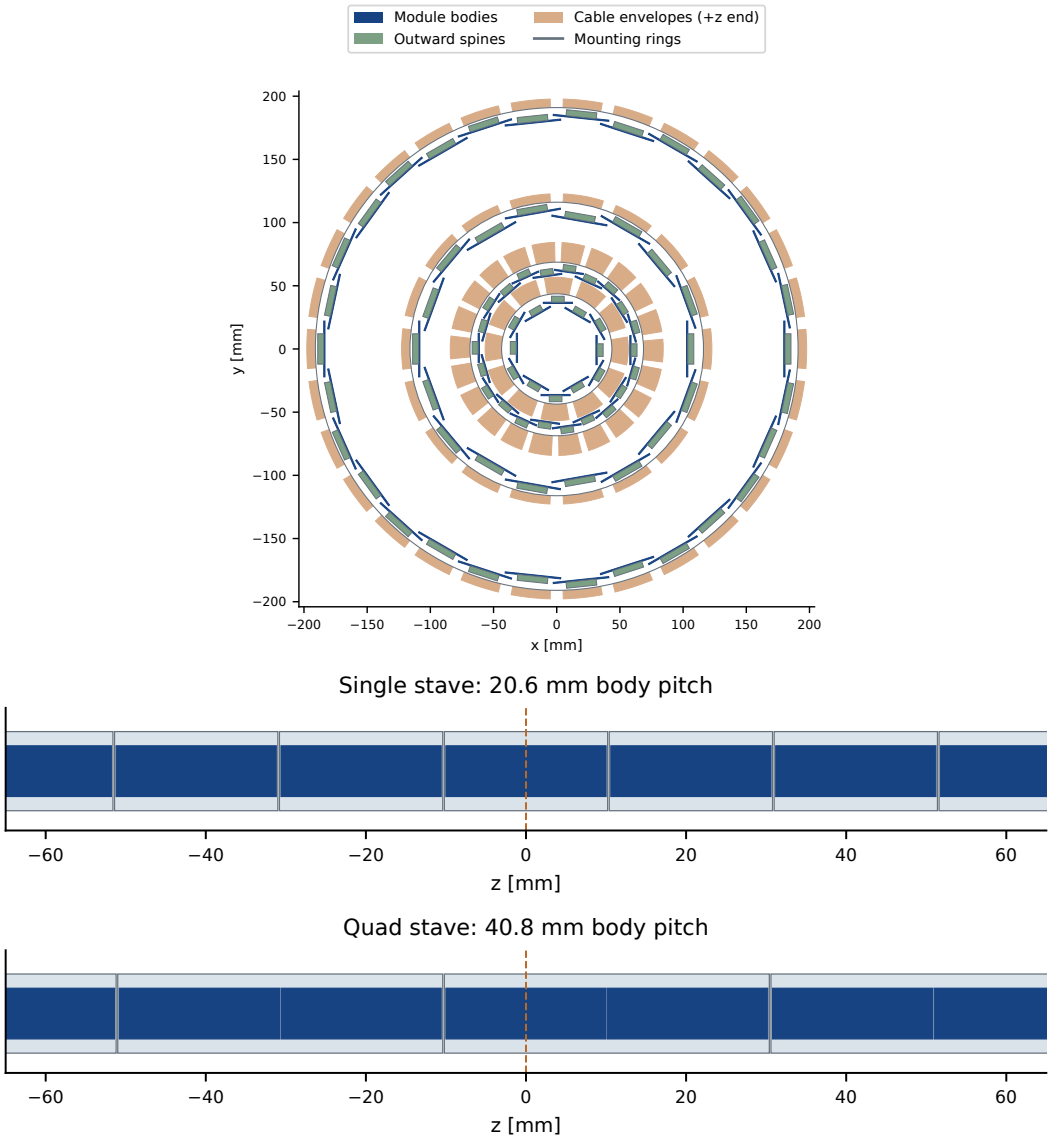


Figure 1.3. Transverse barrel assembly projection (top) and central single/quad row packing (bottom). Module bodies overlap tangentially on staggered radial levels; cable envelopes pass outside the bearing rings. The lower panel resolves active intervals, guards and gaps about $z = 0$.

1.1.3 Endcap layout

The six-ring pattern contains 102 singles and 50 quads, covering eight radial chip rows. Smaller modules follow the inner curvature; quads share support and service interfaces at larger radii. The local axes are tangential u and radial v around every disc. Module centres account for their individual mounting face and sensor depth. Table 1.3 gives the nominal ring parameters.

Table 1.3. Six-ring source pattern. Radii are the uncorrected radial placement parameters; actual module centres retain the face/height-dependent transverse compensation. Each ring occupies both mounting faces and two height levels per face. Quad rings contain two radial chip rows.

Ring	Family	Nominal r [mm]	Modules	Chips
1	Single	41.150	18	18
2	Single	59.283	22	22
3	Single	77.500	28	28
4	Single	95.779	34	34
5	Quad	124.000	22	88
6	Quad	161.000	28	112

The station inventory in Table 1.4 specifies the retained inner rings and both outer quad rings at each reference z position. The progressively larger active aperture follows the first outward half-turn of charged tracks in vacuum, with $|\eta| \leq 4$, $p_{\mathrm{T}} \geq 1\,\mathrm{GeV}$, luminous vertices $|z_0| \leq 150\,\mathrm{mm}$ and $|x_0|, |y_0| \leq 1\,\mathrm{mm}$, and a uniform axial field with $|B_z| \leq 4\,\mathrm{T}$. Both endcaps contain 2 312 modules and 5 012 chip patches. The negative end uses corresponding signed datums and support-facing module frames.

Table 1.4. Selected disc support reference z positions and retained inventory on one side. Ring numbers refer to the six-ring source pattern; quad rings 5 and 6 remain at every station. The negative side uses the opposite datum sign and the same populations. Positions refer to the central support planes; individual sensor planes include the face and height offsets.

Disc	$\lvert z\rvert$ [mm]	Single rings	Singles	Quads	Chips	Circuits
1	615	1, 2, 3, 4	102	50	302	16
2	795	1, 2, 3, 4	102	50	302	16
3	1046	1, 2, 3, 4	102	50	302	16
4	1333	1, 2, 3, 4	102	50	302	16
5	1645	2, 3, 4	84	50	284	14
6	1977	2, 3, 4	84	50	284	14
7	2328	3, 4	62	50	262	12
8	2692	4	34	50	234	10
9	3070	4	34	50	234	10

Modules alternate between the two plate faces and between two heights on each face, keeping overlapping occupied bodies separated in depth. A quad's four patches share one sensor plane. Sensor centres are at local $w = \pm 4.10$ or $\pm 5.75\,\mathrm{mm}$; ASICs and graphite contacts face the plate. An outgoing track crosses the plate before an outward-facing sensor. The first and last discs are compared in Figure 1.4.

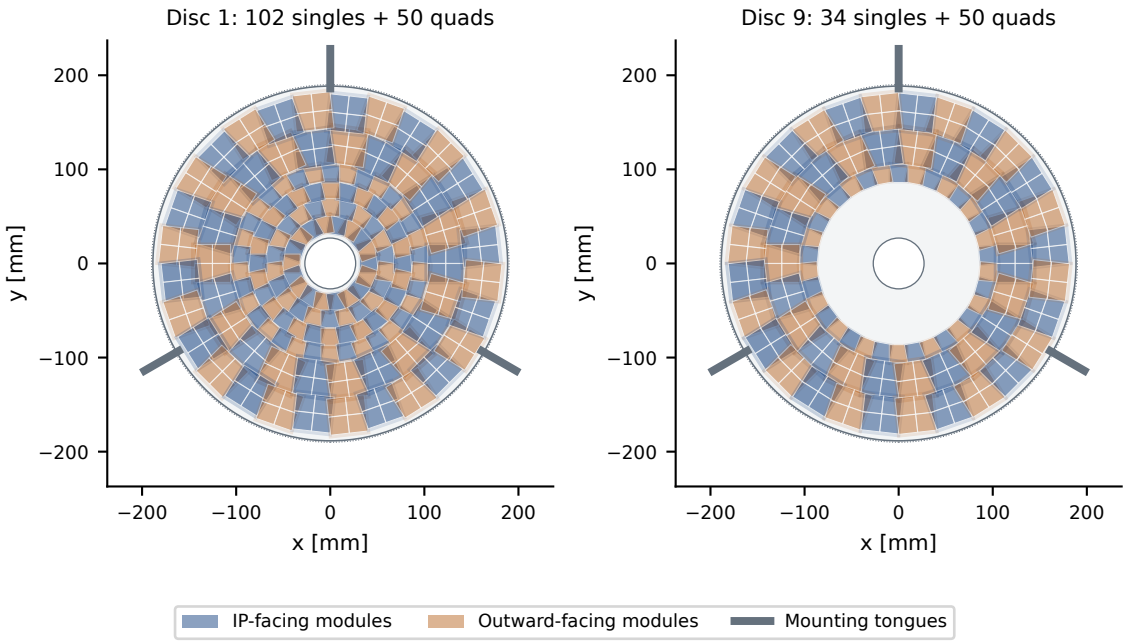


Figure 1.4. First and last positive discs at a common scale. The first has four single rings and two quad rings; the last retains single ring 4 and both quad rings. Colour distinguishes face and height. The views project all sensor depths onto the transverse plane.

1.1.4 Supports and cooling

Barrel staves use carbon-foam cores, CFRP skins, graphite spreaders and two embedded titanium tubes. The 10 mm single and 24 mm quad spines support wider module-facing plates, bringing heat from the chip area toward the cooled core. The support lies radially outside the sensor. Six bearing rings per layer contact the stave backs through discrete CFRP feet at $z = \pm 110, \pm 330, \pm 546$ mm, giving 24 rings and 492 feet. The station at $z = -546$ mm sets the axial datum; the remaining contacts propose longitudinal sliding or compliance during contraction. The rings surround the staves and stiffen the common cage. The closed load path continues through the cage and end connections to the external support; those parts and their stiffness are not specified by the ring/foot solid model. Consequently the approximately 220 mm ring spacing cannot be treated as a set of independently grounded beam supports. Two independent evaporators per stave run in opposite directions between its ends. Carbon sandwiches and titanium cooling tubes have public pixel stave precedents.[3] Figure 1.5 distinguishes the modeled contact geometry from the proposed cage and joints.

The local bending screen uses an assumed axial laminate modulus of 70 – 140 GPa; its proposed limits are 50 µm static sag and 5 µm change during steady running. These are design screens, not measured performance. Sizing the cage requires its laminate section, joints, actual support reactions, service loads and thermal/pressure load cases, including core shear and torsion. A skin thickness or cage wall cannot be selected from ring spacing alone.

Each endcap plate remains a full annulus from $r = 27$ to 188.5 mm, including on discs with fewer active rings. These pixel component dimensions refer to standalone DES-019. The combined DES-025 assembly opens affected passive apertures to 28.8 mm for the 27.8 mm beam-pipe outer wall; sensitive geometry is unchanged. A per-chip pickup

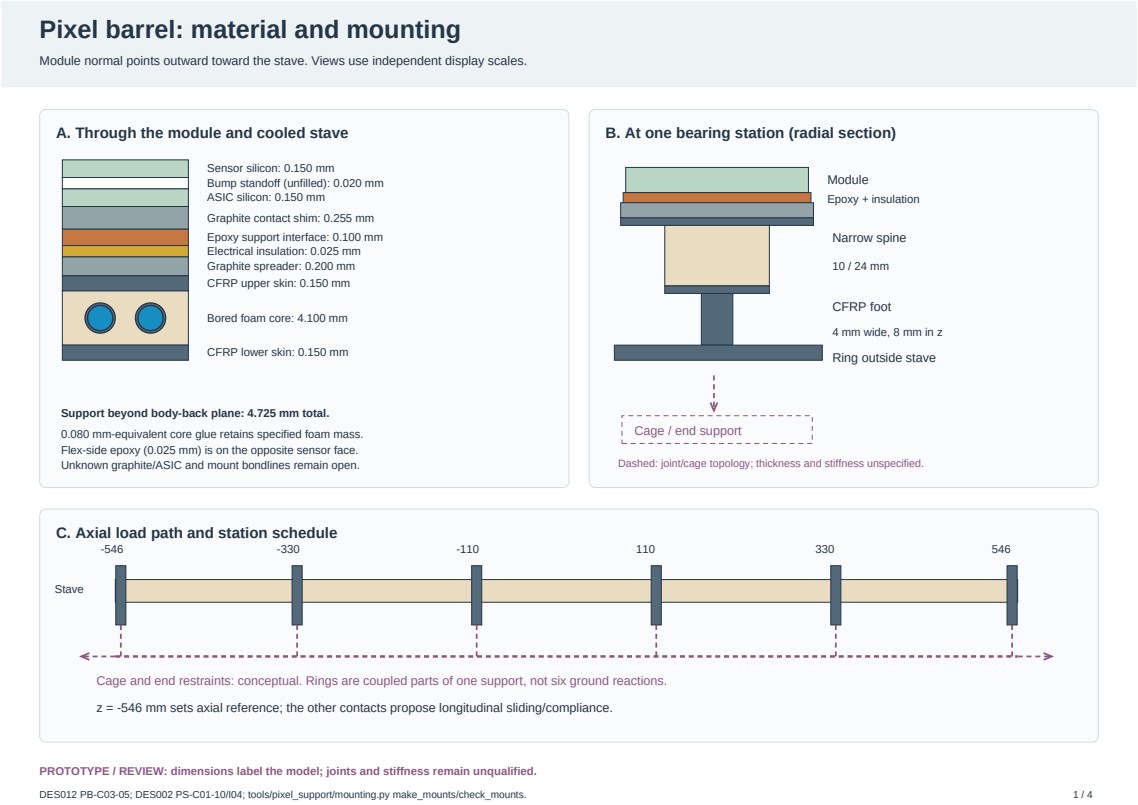


Figure 1.5. Pixel stave material and mounting topology. The module-to-stave interface includes explicit epoxy and electrical insulation. The lower view locates all six ring stations. Dashed cage/end connections are conceptual and carry no modeled thickness or demonstrated stiffness. Thin layers have separate display scales; the labeled thicknesses govern.

and short graphite support foot connect each ASIC to the sandwich and titanium tube. Graphite orientation defines the thermal-conduction path.[4, 6] Azimuthal and radial tube routes occupy separate planes at $w = \pm 1.5\,\mathrm{mm}$. Each chip row has two independent half-ring circuits; the station totals are in Table 1.4, giving 248 circuits across the two endcaps. Three rim tongues transfer loads through proposed cone/slot/plane contacts to box rails on a common CFRP carrier. These tongues are radial extensions of the sandwich, not coolant feeds. The raised graphite foot is a solid thermal spacer for a higher module; it carries no coolant. Its stem is 1.65 mm high on raised modules and vanishes on low modules. Figure 1.7 shows both the local thermal contact and the load path to the carrier. Table 1.5 and Figure 1.6 specify the local passive geometry.

The stave's 0.100 mm epoxy interface is explicit material; the additional 0.080 mm-equivalent internal adhesive is homogenized into the bored core while retaining the specified foam mass. The module flex separately contains 25 µm epoxy. The disc pickup includes 25 µm adhesive and 25 µm thermal-interface material. These allocations do not specify every physical bondline: graphite-to-ASIC and mount-joint bonds, fasteners and their contact resistances require an engineering inventory.

The barrel and endcap coolant temperatures are separate screening assumptions. The barrel's −35 °C input is paired with a prospective $15\,\mathrm{K\,cm^2\,W^{-1}}$ thermal figure of merit; at the $1.05\,\mathrm{W\,cm^{-2}}$ stress heat flux this would give 15.75 K sensor rise. The endcap studies use −40 °C with a different, narrow anisotropic graphite path and retain −35 °C as a warm-coolant sensitivity that fails in recorded cases. The colder reference does not

Table 1.5. Support and mounting ledger. Lengths are in mm. Internal adhesive and inserts replace core material within the stated envelope.

Component	Selected dimensions
Stave stack, module outward	Epoxy interface 0.100; insulation 0.025; graphite 0.200; CFRP 0.150; foam 4.100; CFRP 0.150; total 4.725. Internal adhesive equivalent 0.080.
Barrel bearing rings/feet	Ring axial width 8, radial depth 1; inner clearance 0.5 beyond stave corner. Foot tangential width 4, axial length 8; height follows stagger.
Passive stave span	Singles: $z = -551.0$ to $+552.8$; quads: $z = \pm 552.8$.
Disc sandwich	Core 6.0; each CFRP skin0.15; total 6.3.
Per-chip pickup	Area 384 mm^2; graphite 0.30; cradle 0.075; insulation/interface/adhesive0.025 each; total 0.45.
Disc stem and inserts	Stem section 6×8; heights 0 or 1.65; quad stems shift 1 toward module centre. Core insert 4.50; skin window 0.15; contact area 48 mm^2; adhesive equivalent 0.08.
Disc tongues	Width 10; $r = 186$–228 at $\phi = 90, 210, 330°$.
Carrier rails/shell/flanges	Three 8×8 box rails, wall 0.4 at $r = 223.7$–231.7; shell $r = 231.7$–232.0, $\lvert z \rvert = 609$–3136; end flanges radial 10, axial 2.

Table 1.6. Reference cooling inputs. Diameters/walls are in mm; flow is per independent tube or circuit. The common heat allowance is 2.688 W per chip; the endcap flow budget uses a factor 1.5 stress load.

Location	Ti diameter	Ti wall	CO_2 [°C]	Flow [g s^{-1}]
Inner barrel	2.0	0.11	−35	1.3
Outer barrel	2.8	0.15	−35	2.5
Endcap, inner chip rows	2.8	0.15	−40	1.0
Endcap, outer quad chip rows	2.8	0.15	−40	1.1

establish a requirement for a separate plant or validate the contact model. Measured thermal resistance, irradiated leakage and two-phase hydraulics must close the operating margin; temperature unification needs that analysis.

The barrel tube centres are tangentially at ±2.2 mm for inner staves and ±6 mm for outer staves. The 164 full-length tubes join 82 feed and 82 return branches at each signed end. Thin titanium circuits with evaporative CO_2 provide the reference cooling topology.[5, 6]

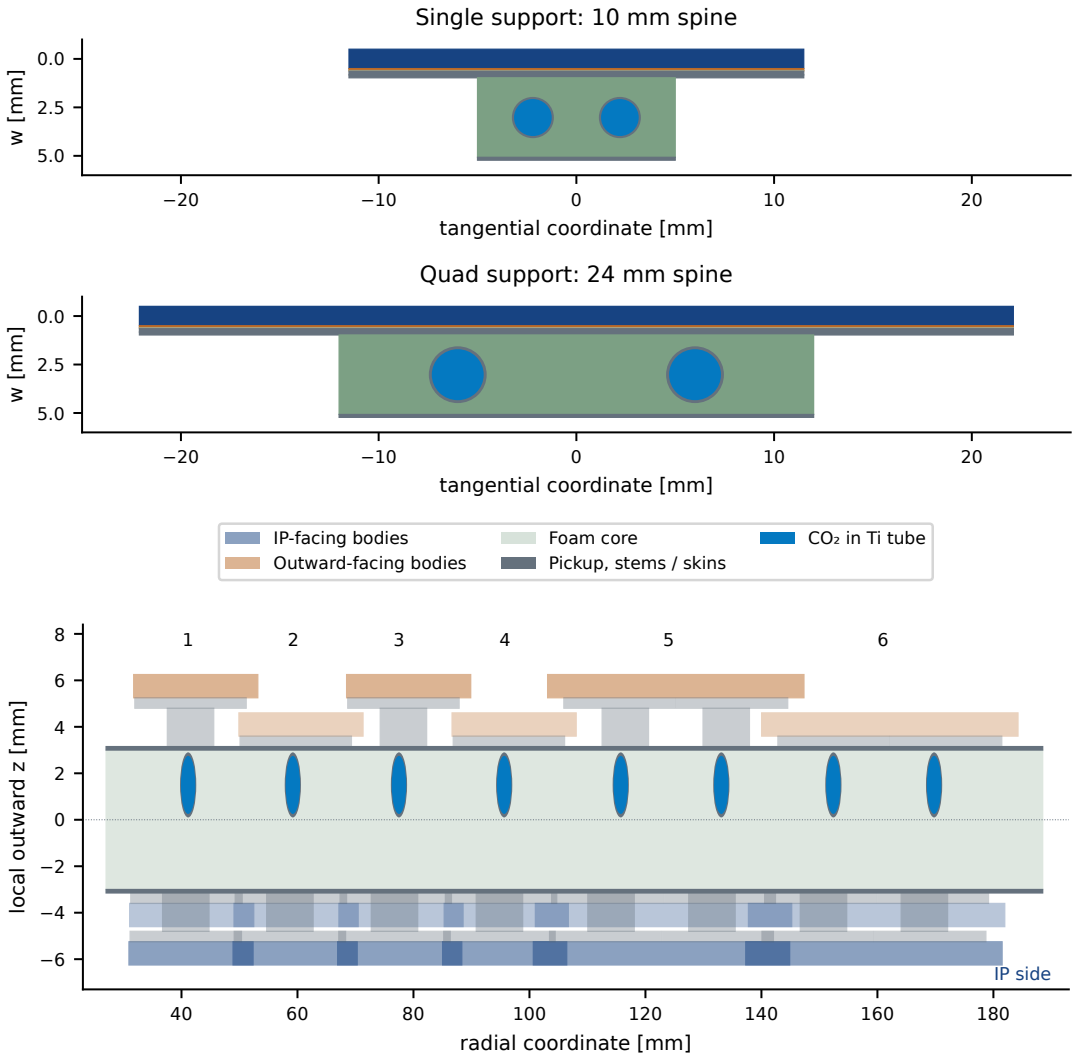


Figure 1.6. Stave sections (top) and a radial/axial disc projection (bottom). Carbon, foam, titanium and coolant form the passive thermal path. The disc panel superposes both mounting faces and heights; short pickups connect the chip backs to the plate. Thin layers are specified in the support ledger.

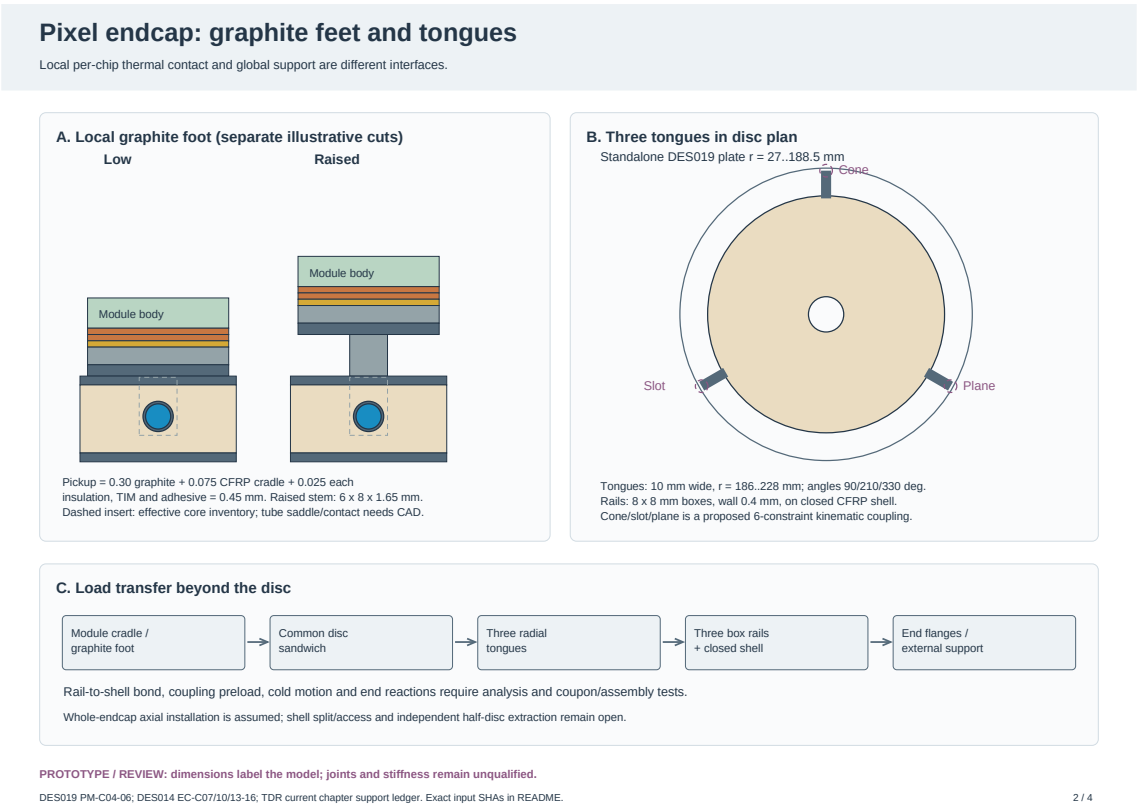


Figure 1.7. Endcap graphite foot and three mounting tongues. The local contact detail separates a low pickup from its raised variant; core insert and tube connections preserve effective material inventory rather than manufacturing connectivity. In plan, three tongues join the disc to rails bonded to the closed carrier shell. Couplings and end connections remain engineering concepts.

1.1.5 Services and material representation

Barrel electrical services follow each half-stave toward its nearest end; the central single-chip module joins the positive half. Two serial chains serve each half-stave, with limits of 16 modules and 32 chips, giving 164 chains at either end. Cable envelopes grow with the cumulative module and chain load and pass outside the bearing rings. Cooling and electrical services collect in twelve sectors in the $|z| = 555$–605 mm bay, then join the axial reservation near $r = 212$ mm. Two sectors serve six staves each and ten serve seven each. Figure 1.8 illustrates the local route; Table 1.7 records its envelope rules. At module level, the proposed pigtail leaves the exposed flex termination beside the ASIC periphery, includes strain relief, and reaches an insulated longitudinal service bundle outside the rings. The pigtail crossing, termination and connector head are conceptual; the current transport cells specify their material/route allocation and do not resolve pinouts, service-loop bends or a continuous cable assembly.

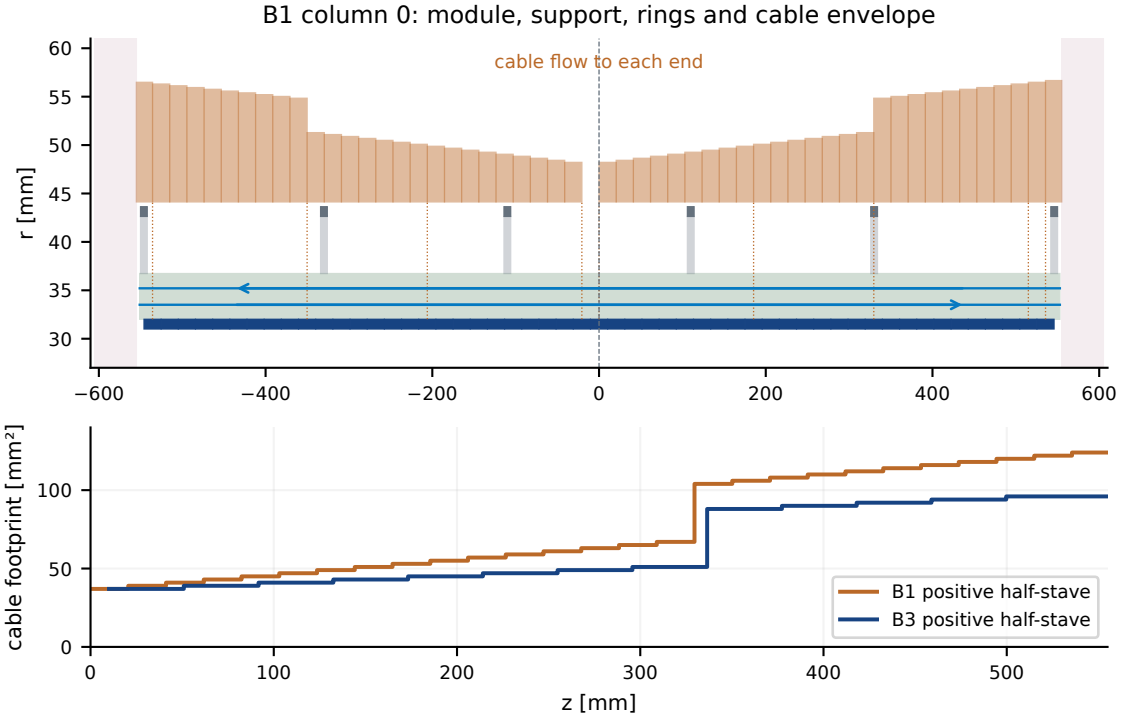


Figure 1.8. Barrel service routes and cumulative cable footprint. The upper panel projects the occupied bodies, support, bearings and conceptual pickups. Counterflow paths are separated radially for display; their physical tubes are separated tangentially in the core.

Endcap tails join ring buses on their own module face. Three radial fans per face gather them at the disc rim. Electrical chains stop at half-ring boundaries and contain at most 32 chips, independently of the cooling circuits: the full disc has 18 chains, reducing to 16, 14 and 12 on the forward stations. A 50 mm-deep collector carries each load into the axial trunk. The last disc uses an inner bypass to the rear turn at $|z| = 3\,150$ mm. Figures 1.10 and 1.11 show the face routes and effective transport cells.

On a disc the pigtail drops through a side slot beside its own graphite foot to a bus on the same mounting face. Ring buses feed the three radial fans on that face, then turn at the rim into the collector and axial trunk. This avoids an assumed pigtail through an opposite-face module. Slots, bond pads, connector access and the rim bend are proposed interfaces;

the present effective service material does not prove those connections mechanically or electrically. Figure 1.9 shows the intended handoffs and their modeling boundaries.

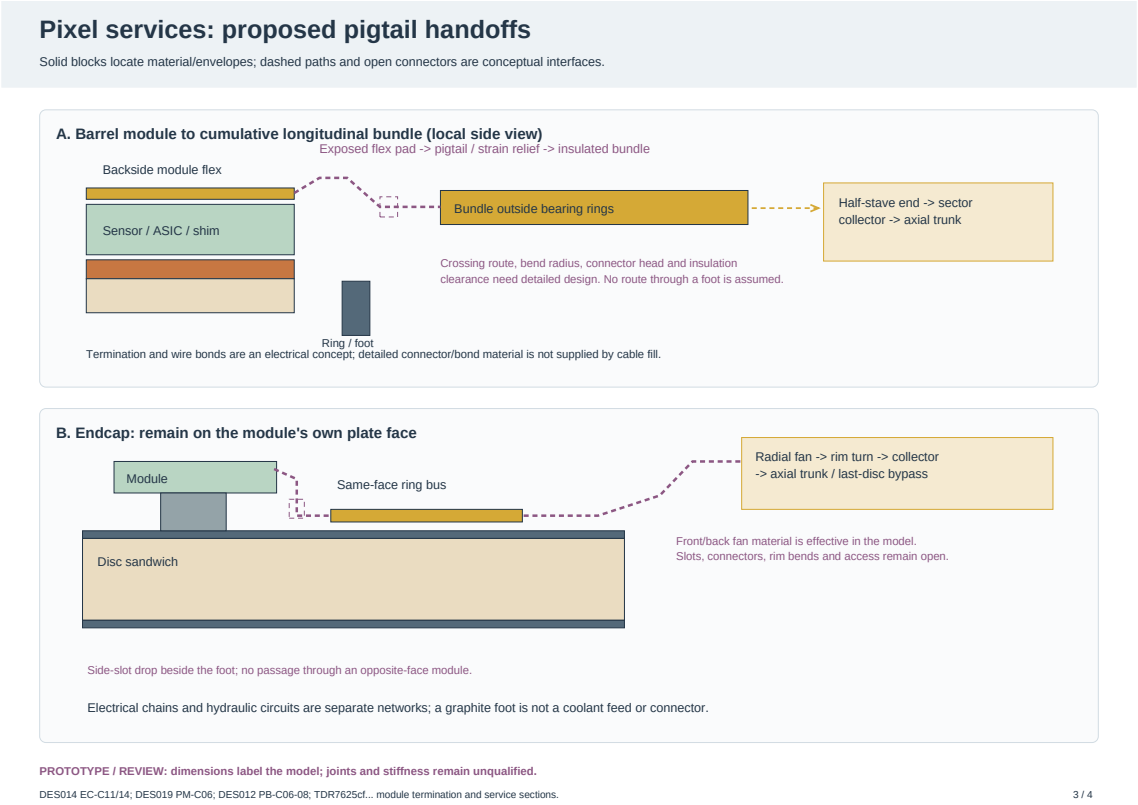


Figure 1.9. Proposed pigtail handoffs. Solid blocks locate modeled material or effective inventory; dashed paths and connector symbols explain the intended electrical/assembly interfaces. Barrel pickups reach the bundle outside the bearing rings; endcap pickups remain on their own plate face before the rim handoff. No connector, strain relief or swept bend is qualified by this drawing.

The simulation separates sensitive silicon from passive module bodies, supports and services. Explicit sensor substrates, chip patches, ASICs, graphite contacts, stave spines, tubes, rings and feet preserve material ordering around the measurements. Effective components conserve their constituent inventory as summarized in Table 1.8.

Remote cable footprint uses provisional volume fractions of 10 % copper, 40 % polyimide and 50 % air, with a separate 50 % inter-cable packing allowance. CO_2 inventory uses a reference density of 1.096 4 g cm^{-3}, the saturated liquid density at −35 °C.[7] The model-inventory mass is 202.753 kg, including 159.383 kg in downstream transport cells. The effective compositions, cooling inputs and mounting contacts specify the simulation baseline and its engineering reference assumptions.

Table 1.7. Standalone DES-019 electrical and remote-service ledger; the combined DES-025 passive-aperture amendment is described with the support geometry. Lengths are in mm, areas in mm^2. Cable footprint and inter-cable packing are separate budgets.

Component	Allocation and route
Barrel cables	One data and one command link per module, footprint 1 each; ancillary footprint 35 per chain from its first pickup. Sector width 75% of stave pitch, packing 50%; inner radius 0.5 beyond the outer bearing ring.
Half-stave terminal loads	Singles: ancillary 70 plus links 52 at $-z$,54 at $+z$; quads: ancillary 70 plus links 26.
Grouped coolant transport	Feed and return branches reserve diameter4; a grouped leg for N staves has envelope diameter $4\sqrt{N}$.
Endcap face flex	Ring buses 8×0.20; module tails 8×0.12; three radial fans per face.
Disc collector	r = 27–190; local outward z = 11.7–61.7 from its datum.
Axial trunk	Reservation r = 190–234; material cells r = 192–231.7. Three 90° sectors separated by 30° mounting sectors centred at $\phi = 90, 210, 330°$. Barrel load enters at $\lvert z\rvert$ = 605; disc 1–8 loads enter after their collectors.
Rear interfaces	Last collector ends at $\lvert z\rvert$ = 3131.7; bypass r = 27–188 to rear turn $\lvert z\rvert$ = 3150. Rear cell extends to $\lvert z\rvert$ = 3300, r = 680. Flange necks at $\lvert z\rvert$ = 609–611 and 3134–3136 have outer radius 222.

Table 1.8. Selected representations and the physical quantities they preserve. The effective components remain identifiable material allocations.

Component	Representation and scope
Disc sandwich	Explicit annulus and disjoint tongues; constituent-normalized core for inserts, adhesive and closeouts.
Cooling	Explicit Ti/CO_2 evaporator tori; summed half-ring length preserved. Radial legs and bends retain effective inventory.
Module pickup	Explicit per-chip pickup stacks and short stems; contact windows and cradle openings preserved through effective volumes.
Disc flex and fittings	Local effective annulus beyond module envelopes; flex, clip, coupling and bend allowances remain separate inventory terms.
Collector and trunk	Effective transport cells preserve conductor, insulation, titanium and coolant volumes through service handoffs.
Carrier	Shell, curved box rails and flanges partitioned to avoid counting intersections twice; coupling pockets remain reserved.

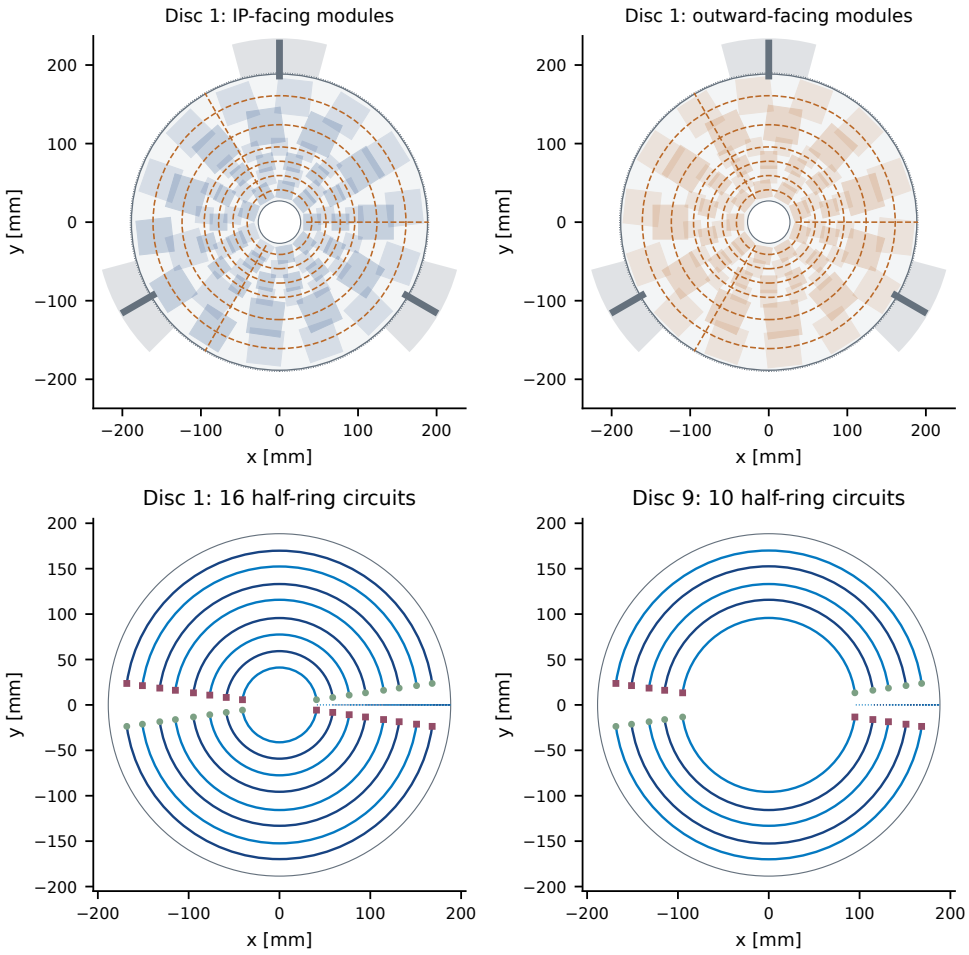


Figure 1.10. Disc face buses and independent half-ring cooling paths. The first/last discs have sixteen/ten cooling circuits. Module outlines preserve their frames; bus lines, fans and inlet/outlet symbols show service topology.

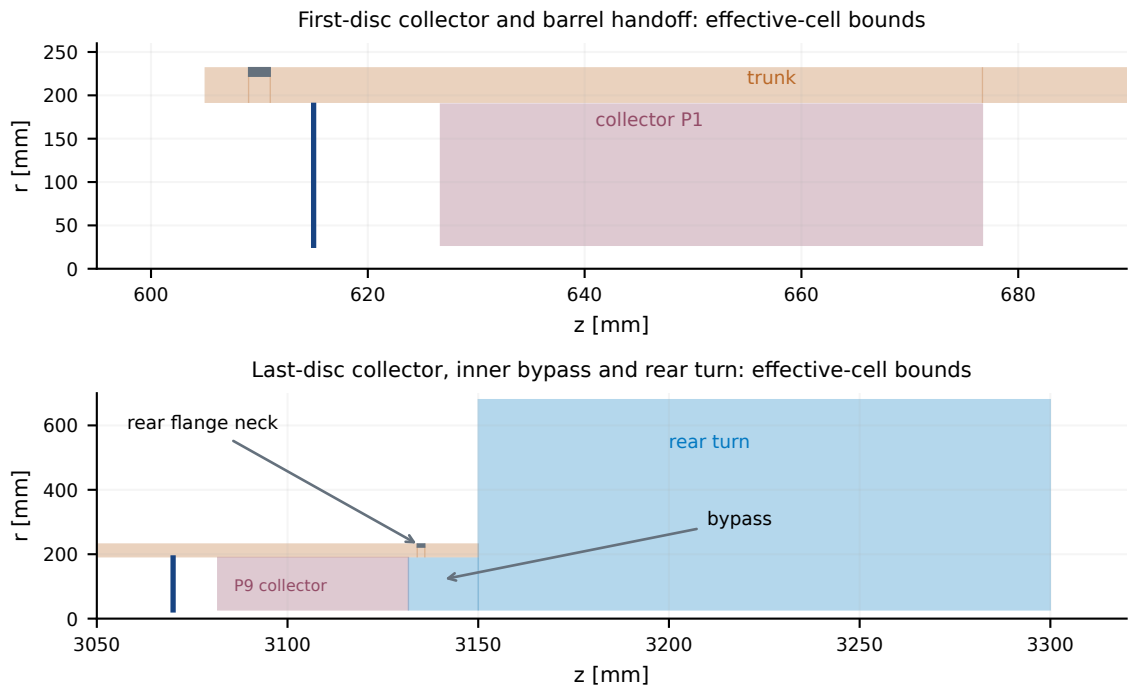


Figure 1.11. Effective collection, trunk, final-disc bypass and rear-turn cells. Carrier-flange necks retain their fixed bounds. Each cell contains the normalized constituent inventory associated with its service handoff.

Pixel detector references

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1.2 Strip mounting interfaces

The cooled carbon sandwich carries the silicon, local electronics and services in both strip systems. In the short-strip system the modules sit on one face through graphite pickups. In the long-strip system the two sensors of each stereo pair attach to opposite faces of the same structural sandwich. Raising a pair increases both outward pickup heights while leaving that common carrier in place. Figure 1.12 follows these local loads through the bearing or petal contacts to their external support interfaces.

Table 1.9. Strip mount topology. Contact dimensions specify the prototype material and clearance fixtures; clamp preload, bond strength, sliding friction and global frame stiffness require engineering qualification.

Assembly	Local load path	Constraints and external interface
Short barrel	Sensor/readout stack → two graphite pickups → 52 mm sandwich → bare back-skin land → 2 mm beveled web → inner bearing ring.	Nine ring stations at $z = 0, \pm 250, \pm 550, \pm 850, \pm 1150$ mm. Centre sets z; other stations propose longitudinal sliding/compliance. Web lies between insulated service routes; its bond/clamp and the ring-to-cage connection remain open. The ring lies at reference radius $r_0 - 20$ to $r_0 - 17$ mm, inside the stave.
Short endcap	Module → two graphite pickups → common petal → three back lugs → inner/outer CFRP frames.	One inner datum and two outer constrained/sliding seats. The frames require a cage connection; cage and detailed fasteners are outside the local model.
Long barrel	Both sensor faces → module glue/spreaders and pickups → common 124 mm sandwich → potted epoxy/Ti lands → keyed edge web and tangent shoe → inner bearing ring.	Seventeen realized stations, pitch 163.75 mm, from $z = -1310$ to +1310 mm. Centre sets z; others reserve ±0.3 mm longitudinal motion. Ground-ring reactions and joint compliance are assumptions of the local screen.
Long endcap	Both faces → pickups → one petal sandwich with integral webs → potted/keyed lands → two outer contacts and one inner key → bearing rings.	One outer contact fixes in-plane position, the other slides tangentially; the inner key slides radially. Proposed travel is ±0.3 mm. Ring/cage reactions and the narrow inner key need plate/joint analysis.

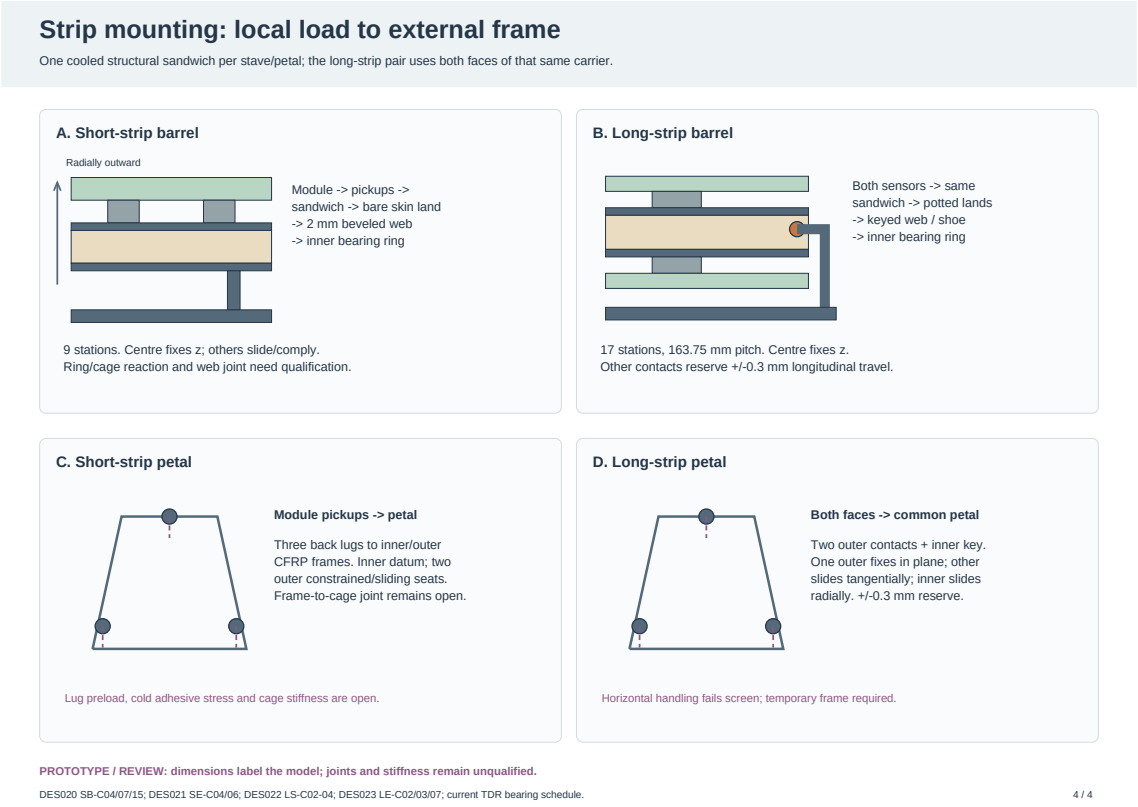


Figure 1.12. Strip load paths and proposed constraint scheme. Short strips use single-face module pickups; long strips place both sensor faces on one cooled structural sandwich. Barrel contacts transfer load at discrete bearing stations; petals use three seats. Dashed links mark the external frame and unresolved joint interfaces. Drawing positions are schematic; the chapter parameter tables specify the actual dimensions.

The mounting schemes describe simulation geometry and engineering hypotheses. The long-barrel gravity screen includes core shear and a separate joint allocation; it assumes stiff bearing reactions. The long-petal horizontal handling case exceeds its displacement screen, requiring a temporary handling frame and plate/joint/global-ring analysis. Adhesive stresses during cooldown, torsion, local load introduction, service forces and vibration remain open across the four assemblies. A successful geometry construction or overlap check does not qualify these mechanical interfaces.